

Introduction to Lie Group Theory

Tejaswi
Supervisor : Prof. Anupam Singh

Summer 2026

Contents

1	Matrix Groups	1
1.1	Groups of Matrices	1
1.2	Matrix Groups	2
1.3	Complex Matrices as Real Matrices	5
1.4	Matrix Groups of Normed Vector Spaces	5
1.5	Matrix Exponential and Logarithm	6
1.6	Differential Equations in Matrices	7
1.7	One Parameter Subgroups in Matrix Groups	8
2	Algebras and Quaternions	10
3	Lie Algebras and Tangent Spaces	11
4	Lie Groups	12
5	Homogenous Spaces	13

Chapter 1

Matrix Groups

1.1 Groups of Matrices

Let $M_{m,n}(\mathbb{K})$ be the set of $m \times n$ matrices whose entries are in $\mathbb{K}$, a commutative field. We will also use the special notations $M_n(\mathbb{K}) = M_{n,n}(\mathbb{K})$, and $\mathbb{K}^n = M_{n,1}(\mathbb{K})$.

$M_{m,n}(\mathbb{K})$ is a vector space over $\mathbb{K}$ with dimension mn . As well as being a vector space over $\mathbb{K}$ with dimension n^2 , $M_n(\mathbb{K})$ is also a ring with the usual addition and multiplication of matrices. It is also an important example of a finite dimensional algebra over $\mathbb{K}$.

We will use the notation

$$GL_n(\mathbb{K}) = \{A \in M_n(\mathbb{K}) : \det A \neq 0\}$$

for the set of invertible $n \times n$ matrices, known as the $n \times n$ *general linear group*, and

$$SL_n(\mathbb{K}) = \{A \in M_n(\mathbb{K}) : \det A = 1\} \subseteq GL_n(\mathbb{K})$$

for the set of $n \times n$ unimodular matrices, known as the $n \times n$ *special linear group*.

Theorem 1.1. *The sets $GL_n(\mathbb{K})$ and $SL_n(\mathbb{K})$ are groups under matrix multiplication. Furthermore, $SL_n(\mathbb{K})$ is a subgroup of $GL_n(\mathbb{K})$.*

Since determinant of a matrix is a polynomial in the entries of the matrix, it can be proved that the determinant function

$$\det : M_n(\mathbb{K}) \longrightarrow \mathbb{K}$$

is continuous.

Proposition 1.1. Let $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. Then

- (i) $GL_n(\mathbb{K}) \subseteq M_n(\mathbb{K})$ is an open subset;
- (ii) $SL_n(\mathbb{K}) \subseteq M_n(\mathbb{K})$ is a closed subset.

Proof. $GL_n(\mathbb{K}) = M_n(\mathbb{K}) - \det^{-1}\{0\}$, which is open since $\{0\}$ is closed. Similarly, $SL_n(\mathbb{K}) = \det^{-1}\{1\}$, which is closed since the singleton $\{1\}$ is closed in $\mathbb{K}$. ■

Definition 1.1. Let G be a topological space, and view $G \times G$ as the product space. Suppose that G is also a group with multiplication map $\text{mult} : G \times G \rightarrow G$, and inverse map $\text{inv} : G \rightarrow G$. G is a *topological group* if mult and inv are continuous.

Theorem 1.2. For $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$, $GL_n(\mathbb{K})$ and $SL_n(\mathbb{K})$ are topological groups.

1.2 Matrix Groups

Definition 1.2. A subgroup G of $GL_n(\mathbb{K})$ is known as a *matrix group*, given it is a closed subset of $GL_n(\mathbb{K})$.

Remark. We may consider $GL_n(\mathbb{K})$ as a subgroup of $GL_{n+1}(\mathbb{K})$ by identifying the $n \times n$ matrix $A = [a_{ij}]$ with

$$\begin{bmatrix} A & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} a_{11} & \cdots & a_{1n} & 0 \\ \vdots & \ddots & \vdots & \vdots \\ a_{n1} & \cdots & a_{nn} & 0 \\ 0 & \cdots & 0 & 1 \end{bmatrix}$$

And it is easily verified that $GL_n(\mathbb{K})$ is closed in $GL_{n+1}(\mathbb{K})$. With the aid of this embedding, any matrix subgroup of $GL_n(\mathbb{K})$ can also be viewed as a matrix subgroup of $GL_{n+1}(\mathbb{K})$.

Definition 1.3. Let G, H be two matrix groups. A group homomorphism $\varphi : G \rightarrow H$ is a *continuous homomorphism of matrix groups* if it is continuous and its image $\varphi(G)$ is a closed subgroup of H .

Examples. Some important examples of real and complex matrix groups:

1. Groups of Upper Triangular Matrices

For $n \geq 1$, an $n \times n$ matrix $A = [a_{ij}]$ is said to be *upper triangular* if $a_{ij} = 0$ if $i < j$. An upper triangular matrix is said to be *unipotent* if all of its diagonal entries are 1.

The *upper triangular subgroup* or *Borel subgroup* of $GL_n(\mathbb{K})$ is

$$UT_n(\mathbb{K}) = \{A \in GL_n(\mathbb{K}) : A \text{ is upper triangular}\},$$

while the *unipotent subgroup* of $GL_n(\mathbb{K})$ is

$$SUT_n(\mathbb{K}) = \{A \in GL_n(\mathbb{K}) : A \text{ is unipotent}\}.$$

It is easy to verify that $UT_n(\mathbb{K})$ and $SUT_n(\mathbb{K})$ are closed subgroups of $GL_n(\mathbb{K})$, and that $SUT_n(\mathbb{K})$ is a subgroup of $UT_n(\mathbb{K})$.

2. Affine Groups

The *n-dimensional affine group* over $\mathbb{K}$ is

$$Aff_n(\mathbb{K}) = \left\{ \begin{bmatrix} A & t \\ 0 & 1 \end{bmatrix} : A \in GL_n(\mathbb{K}), t \in \mathbb{K}^n \right\}$$

which is clearly a closed subgroup of $GL_{n+1}(\mathbb{K})$. If we identify $x \in \mathbb{K}^n$ with $\begin{bmatrix} x \\ 1 \end{bmatrix} \in \mathbb{K}^{n+1}$, then as a consequence of

$$\begin{bmatrix} A & t \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ 1 \end{bmatrix} = \begin{bmatrix} Ax + t \\ 1 \end{bmatrix}$$

we obtain an action of $Aff_n(\mathbb{K})$ on $\mathbb{K}^n$.

The group $Aff_n(\mathbb{K})$ can be expressed as the semi-direct product of $Trans_n(\mathbb{K})$ and $GL_n(\mathbb{K})$, where

$$Trans_n(\mathbb{K}) = \left\{ \begin{bmatrix} I_n & t \\ 0 & 1 \end{bmatrix} : t \in \mathbb{K}^n \right\}$$

is a closed normal subgroup of $Aff_n(\mathbb{K})$. That is,

$$Aff_n(\mathbb{K}) = GL_n(\mathbb{K}) \ltimes Trans_n(\mathbb{K}).$$

3. Orthogonal Groups

The *n × n real orthogonal group* is the subset

$$O(n) = \{A \in GL_n(\mathbb{R}) : A^T A = I_n\},$$

where A^T is the transpose of $A = [a_{ij}]$ whose entries are given by $(A^T)_{ij} = a_{ji}$. $O(n)$ is a closed subgroup of $GL_n(\mathbb{R})$. For all $A \in O(n)$, since $\det A = \det A^T$, $\det A = \pm 1$. The subgroup

$$SO(n) = \{A \in O(n) : \det A = 1\}$$

is known as the *special orthogonal group*. One of the main reasons for the study of orthogonal groups is due to their relation with isometries. If $A \in GL_n(\mathbb{R})$, then it can be proved that A is a linear isometry *iff* A is orthogonal. The elements of $SO(n)$ are called *direct isometries* or *rotations*.

4. Unitary Groups

The $n \times n$ *unitary group* is the subset

$$U(n) = \{A \in GL_n(\mathbb{C}) : A^*A = I_n\},$$

where A^* is the *hermitian conjugate* of A , that is $A^*)_{ij} = a_{ji}$. $U(n)$ is clearly a closed subgroup of $GL_n(\mathbb{C})$. The *special unitary group* is the subset

$$SU(n) = \{A \in U(n) : \det A = 1\},$$

which is also a normal subgroup of $U(n)$.

5. Symplectic Groups

A matrix S is called *skew symmetric* if $S^T = -S$. The standard example of this is built up using the 2×2 block

$$J = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}.$$

When $m \geq 1$ we have the non degenerative skew symmetric matrix

$$J_{2m} = \begin{bmatrix} J & O_2 & \cdots & O_2 \\ O_2 & J & \cdots & O_2 \\ \vdots & \vdots & \ddots & \vdots \\ O_2 & O_2 & \cdots & J \end{bmatrix}.$$

The matrix group

$$\text{Symp}_{2m}(\mathbb{R}) = \{A \in GL_{2m}(\mathbb{R}) : A^T J_{2m} A = J_{2m}\}$$

is called the $2m \times 2m$ *real symplectic group*. It can be proved that if $A \in \text{Symp}_{2m}(\mathbb{R})$, then $\det A = 1$.

1.3 Complex Matrices as Real Matrices

Similar to how complex numbers can be viewed as a 2-dimensional real vector space with basis $(1, i)$, every $n \times n$ complex matrix can also be viewed as a $2n \times 2n$ real matrix using the following construction or the ones similar to it.

We identify each complex number $z = x + iy$ with a 2×2 matrix by defining a function

$$\rho : \mathbb{C} \longrightarrow M_2(\mathbb{R}); \quad \rho(x + iy) = \begin{bmatrix} x & -y \\ y & x \end{bmatrix}.$$

This can also be expressed as

$$\rho(x + iy) = xI_2 - yJ.$$

As ρ is an injective ring homomorphism, we can view $\mathbb{C}$ as a subring of $M_2(\mathbb{R})$. Notice that $\rho(\bar{z}) = \rho(z)^T$.

For a $n \times n$ complex matrix Z , we define a function $\rho_n : M_n(\mathbb{C}) \longrightarrow M_{2n}(\mathbb{R})$;

$$\rho_n(Z) = \begin{bmatrix} \rho(z_{11}) & \rho(z_{12}) & \cdots & \rho(z_{1n}) \\ \rho(z_{21}) & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \vdots \\ \rho(z_{n1}) & \cdots & \cdots & \rho(z_{nn}) \end{bmatrix}$$

Then ρ_n is an continuous injective ring homomorphism. Generally for $n \times n$ real matrices X and Y ,

$$\rho_n(X + iY) = \rho_n(X) - J_{2n}\rho_n(Y)J_{2n}.$$

1.4 Matrix Groups of Normed Vector Spaces

It is often desirable to generalize the notion of matrix spaces to arbitrary finite dimensional real or complex vector spaces. Here we assume $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$.

Theorem 1.3. *Let (U, μ) and (V, ν) be two finite dimensional normed vector spaces of the same dimension over $\mathbb{K}$. Then any linear isomorphism $\varphi : U \longrightarrow V$ is a homeomorphism.*

Corollary. *If μ and ν are two norms on a finite dimensional vector space, then they give rise to the same topology.*

Proof. The identity function is a linear isomorphism, which must be continuous. This implies that every open set in the topology of ν is open in the topology of μ and vice-versa. ■

Given a finite dimensional normed vector space V , ν and a linear transformation $\alpha : V \longrightarrow V$, we define the *operator norm* of α with respect to ν by

$$\|\alpha\|_\nu = \sup\{\nu(\alpha(x)) : x \in V, \nu(x) = 1\}$$

This defines a norm on $\text{End}_{\mathbb{K}}(V)$, the set of all linear transformations $V \longrightarrow V$, which makes it a finite dimensional normed vector space over $\mathbb{K}$.

Proposition 1.2. *If μ and ν are two norms on a finite dimensional vector space V , then the topologies associated to $\text{End}_{\mathbb{K}}(V)$ with respect to μ and ν are the same.*

Inside $\text{End}_{\mathbb{K}}(V)$ we have the group of linear isomorphisms, $\text{GL}_{\mathbb{K}}(V) \subseteq \text{End}_{\mathbb{K}}(V)$, the *general linear group* of the vector space V .

Proposition 1.3. *Let U and V be two vector spaces of the same dimension. If $\varphi : U \longrightarrow V$ is a linear isomorphism, then it induces a continuous group isomorphism with a continuous inverse,*

$$\varphi_* : \text{GL}_{\mathbb{K}}(U) \longrightarrow \text{GL}_{\mathbb{K}}(V); \quad \varphi_*(\alpha) = \varphi \circ \alpha \circ \varphi^{-1}$$

Generally, given a finite basis for V , there is an associated linear isomorphism $V \cong \mathbb{K}^n$ for some n , a ring isomorphism $\text{End}_{\mathbb{K}}(V) \cong M_n(\mathbb{K})$ and a group isomorphism $\text{GL}_{\mathbb{K}}(V) \cong \text{GL}_n(\mathbb{K})$.

Theorem 1.4. *Let V be a finite dimensional vector space over $\mathbb{K}$ with dimension n and let $\varphi : V \longrightarrow \mathbb{K}^n$ be a linear isomorphism. Then $\varphi_* : \text{GL}_{\mathbb{K}}(V) \longrightarrow \text{GL}_n(\mathbb{K})$ is a continuous group isomorphism with a continuous inverse. Hence $\text{GL}_{\mathbb{K}}(V)$ is a matrix group.*

1.5 Matrix Exponential and Logarithm

We define the exponential and logarithm functions for matrices using their power series. Throughout this section, we assume $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$.

The *exponential function* is, $\exp : M_n(\mathbb{K}) \longrightarrow \text{GL}_n(\mathbb{K})$;

$$\exp(A) = \sum_{n \geq 0} \frac{1}{n!} A^n = I + A + \frac{1}{2!} A^2 + \frac{1}{3!} A^3 + \dots$$

Proposition 1.4. *Let $A \in M_n(\mathbb{K})$.*

- (i) *For $u, v \in \mathbb{C}$, $\exp((u + v)A) = \exp(uA) \exp(vA)$.*
- (ii) *$\exp(A)^{-1} = \exp(-A)$.*
- (iii) *If $B \in M_n(\mathbb{K})$ commutes with A , then $\exp(A + B) = \exp(A) \exp(B)$.*

We define the logarithm function by

$$\log : N_{M_n(\mathbb{K})}(I; 1) \longrightarrow M_n(\mathbb{K});$$

$$\log(A) = \sum_{n \geq 1} \frac{(-1)^{n-1}}{n} (A - I)^n$$

Proposition 1.5. *The exp and log functions satisfy*

- (i) *if $\|A - I\| < 1$, then $\exp(\log(A)) = A$;*
- (ii) *if $\|\exp(B) - I\| < 1$, then $\log(\exp(B)) = B$.*

The functions exp and log are continuous and in fact infinitely differentiable on their domains. By continuity of exp at O , there is a $\delta_1 > 0$ such that

$$N_{M_n(\mathbb{K})}(O; r) \subseteq \exp^{-1} N_{GL_n(\mathbb{K})}(I; 1).$$

Since $\exp N_{M_n(\mathbb{K})}(O; r) \subseteq N_{M_n(\mathbb{K})}(I; e^r - 1)$, we can take $\delta_1 = \log 2$. Hence we have the following result.

Proposition 1.6. *The exponential function exp is injective when restricted to the open subset $N_{M_n(\mathbb{K})}(O; \log 2) \subseteq M_n(\mathbb{K})$. Hence it is locally a diffeomorphism at O with the local inverse log.*

1.6 Differential Equations in Matrices

Definition 1.4. *A differentiable curve in $M_n(\mathbb{K})$ is a function $\alpha : (a, b) \longrightarrow M_n(\mathbb{K})$ for which the derivative $\alpha'(t)$ exists for each $t \in (a, b)$. Here $\alpha'(t)$ is defined as an element of $M_n(\mathbb{K})$ by*

$$\alpha'(t) = \lim_{s \rightarrow t} \frac{1}{(s - t)} (\alpha(s) - \alpha(t)),$$

provided that the limit exists.

Consider the first order differential equation $\alpha'(t) = \alpha(t)A$, where α is assumed to be a differentiable curve.

Theorem 1.5. For $A, C \in M_n(\mathbb{R})$ with A non-zero, and $a < 0 < b$, the differential equation $\alpha'(t) = \alpha(t)A$, has a unique solution $\alpha : (a, b) \rightarrow M_n(\mathbb{R})$ for which $\alpha(0) = C$. Furthermore, if C is invertible, then so is $\alpha(t)$ for $t \in (a, b)$.

Proof. Consider the function $\alpha : (a, b) \rightarrow M_n(\mathbb{R})$; $\alpha(t) = \exp(tA)$. It is differentiable with

$$\alpha'(t) = \sum_{k \geq 1} k \frac{t^{k-1}}{(k-1)!} A^k = A \exp(tA)$$

Hence α satisfies the differential equation with boundary condition $\alpha(0) = I$. Clearly, $\alpha(t) = C \exp(tA)$ is one solution to $\alpha(0) = C$. If β is another such solution, then $\gamma(t) = \beta(t) \exp(-tA)$ satisfies

$$\begin{aligned} \gamma'(t) &= \beta'(t) \exp(-tA) + \beta(t) \exp(-tA)(-A) \\ &= \beta(t)A \exp(-tA) - \beta(t)A \exp(-tA) \\ &= 0. \end{aligned}$$

Hence γ is a constant function with $\gamma(t) = \gamma(0) = C$. Thus $\beta(t) = C \exp(tA) = \alpha(t)$. ■

1.7 One Parameter Subgroups in Matrix Groups

Definition 1.5. A differentiable curve in G is a function $\gamma : (a, b) \rightarrow G$ for which derivative $\gamma'(t)$ exists at each $t \in (a, b)$.

We usually assume that $a < 0 < b$ when considering such curves. Now suppose that $\varepsilon > 0$ or $\varepsilon = \infty$.

Definition 1.6. A one-parameter subgroup in G is a continuous function $\gamma : (-\varepsilon, \varepsilon) \rightarrow G$ which is differentiable at 0 and also satisfies $\gamma(s + t) = \gamma(s)\gamma(t)$ whenever $s, t, s + t \in (-\varepsilon, \varepsilon)$. If $\varepsilon = \infty$, then $\gamma : \mathbb{R} \rightarrow G$ is called a one-parameter subgroup of G .

Proposition 1.7. Let $\gamma : (-\varepsilon, \varepsilon) \rightarrow G$ be a one-parameter semigroup. Then for every $t \in (-\varepsilon, \varepsilon)$, γ is differentiable at t , and

$$\gamma'(t) = \gamma'(0)\gamma(t) = \gamma(t)\gamma'(0).$$

Proposition 1.8. *Let $\gamma : (-\varepsilon, \varepsilon) \longrightarrow G$ be a one-parameter semigroup. Then there is a unique extension to a one-parameter group $\tilde{\gamma} : \mathbb{R} \longrightarrow G$.*

Proof. Let $t \in \mathbb{R}$. Then for a large enough natural number m , $t/m \in (-\varepsilon, \varepsilon)$, hence $\gamma(t/m), \gamma(t/m)^m \in G$. Similarly for a second natural number n , $\gamma(t/n), \gamma(t/n)^n \in G$. Since $mn \geq m$ and $mn \geq n$, we also have $t/mn \in (-\varepsilon, \varepsilon)$. Therefore,

$$\begin{aligned}\gamma(t/n)^n &= \gamma(mt/mn) \\ &= \gamma(t/mn)^{mn} \\ &= \gamma(nt/mn)^m \\ &= \gamma(t/m)^m\end{aligned}$$

Thus $\gamma(t/n)^n = \gamma(t/m)^m$. This defines a function

$$\tilde{\gamma} : \mathbb{R} \longrightarrow G; \quad \tilde{\gamma}(t) = \gamma(t/n)^n$$

for large n . It is easy to see that $\tilde{\gamma}$ is an one-parameter subgroup in G . ■

Theorem 1.6. *Let $\gamma : \mathbb{R} \longrightarrow G$ be a one-parameter group in G . Then it has the form $\gamma(t) = \exp(tA)$ for some $A \in M_n(\mathbb{K})$.*

Proof. Let $A = \gamma'(0)$. Since γ satisfies the differential equation $\gamma'(t) = \gamma(t)A$, $\gamma(0) = A$, by Theorem 1.5, this equation has a unique solution $\gamma(t) = \exp(tA)$. ■



Chapter 2

Algebras and Quaternions



Chapter 3

Lie Algebras and Tangent Spaces



Chapter 4

Lie Groups



Chapter 5

Homogenous Spaces



References

- [Baker, 2002] Baker, A. (2002). *Matrix Groups*. Springer Undergraduate Mathematics Series. Springer-Verlag.
- [Trapp, 2005] Trapp, K. (2005). *Matrix Groups for Undergraduates*. Student Mathematical Library. American Mathematical Society, second edition.